

SceneMotion 3D Reconstruction Report

SceneMotion-3D estimates relative monocular trajectory by default. Absolute metric scale requires additional information such as known camera intrinsics, known object size, known camera height, stereo baseline, depth sensor, IMU, or ground-truth alignment for evaluation. Fallback pseudo-depth is provided for offline demonstration only and must not be interpreted as metric depth.

Metric	Value
video_path	/mnt/data/SceneMotion-3D/sample_data/videos/synthetic_scene.mp4
number_of_frames	48
extracted_frames	16
selected_keyframes	8
average_keypoints_per_keyframe	1638.125
average_matches_per_pair	532.5714285714286
average_inlier_ratio	0.6680122740205736
valid_pose_pairs	7
rejected_pose_pairs	0
sparse_3d_point_count	3371
mean_reprojection_error	0.0
max_reprojection_error	0.0
trajectory_length_relative_units	7.0
depth_maps_generated	4
dense_cloud_point_count	4240
depth_provider	fallback_pseudo_depth
intrinsics_source	estimated
scale_note	Relative monocular units only; no metric scale claimed.
bundle_adjustment_note	Includes lightweight local refinement plus optional small-window BA modules; not a replacement for mature SfM systems like COLMAP.
scale_mode	relative
scale_source	monocular_relative_units
scale_factor	1.0
scale_confidence	0.3
rejected_pairs_total	0
rejected_low_matches	0
rejected_low_parallax	0
rejected_blur	0

Metric	Value
rejected_dynamic_scene	0
rejected_degenerate_motion	0
loop_candidates	4
loop_candidates_verified	4
timing_profile	{ "frame_extraction_time": 0.149016, "keyframe_selection_time": 0.485727, "feature_extraction_time": 0.163268, "matching_time": 2.189775, "pose_estimation_time": 0.576971, "triangulation_time": 0.09645, "depth_estimation_time": 0.25363, "point_cloud_export_time": 0.098169 }
quality_narrative	{ "reconstruction_quality_grade": "Strong", "tracking_stability": 0.6680122740205736, "scale_mode": "relative", "likely_failure_reasons": [], "recommended_fixes": ["Avoid pure rotation; add translation/parallax.", "Provide a camera calibration JSON.", "Use known scale reference, RGB-D, stereo, or IMU for metric scale."] }

Warnings and limitations

- Camera intrinsics are approximate. Use calibration JSON for better geometry.
- Monocular trajectory uses relative scale only unless external scale is provided.
- Offline-safe pseudo-depth fallback; not metric depth.

Recommended fixes

- Avoid pure rotation; add translation/parallax.
- Provide a camera calibration JSON.
- Use known scale reference, RGB-D, stereo, or IMU for metric scale.